How Chiplet Architecture is Taking Over the Monolithic SoC

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Abstract

The semiconductor industry has long relied on Moore's Law as a primary reason for fitting increasing amounts of transistors into smaller and more miniscule dies. However, the economical limits that the manufacturing process had yet to experience are suddenly a factor now more than ever. Rather than integrating all processor functionalities onto one die, chiplet-based architectures introduced the idea of separating systems into smaller, specialized chips that are created and tested individually while being packaged later. This paper presents an in-depth exploration into the historical forces that drove away monolithic chip designs, the functionality of chiplet systems and what limitations they bring, and the sustainable integration of chiplets with artificial intelligence. Drawing upon recent studies that include cost modeling, carbon efficient solutions, and memory/power optimization strategies, this study evaluates chiplet architecture against concrete conditions with an emphasis on AI acceleration.

Keywords: chiplet architecture, monolithic SoC, semiconductor manufacturing, AI acceleration, sustainable computing

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In the past, the semiconductor industry has strived to make their chips bigger and faster to generate products with superior performance. The belief that a higher density of transistors would lead to quicker operations fueled the monolithic processor chip to become the fundamental unit of computing progress. Moore's Law, an observation regarding transistor usage made by Gordon Moore, was a major contributor to this assumption stating that the number of transistors on a microchip doubles about every two years which leads to an exponential growth in computational power. While this prediction has been followed by many semiconductor companies to help plan for long term goals, fabrication technologies have met physical and economic limits causing transistor growth to decline. Building a monolithic die meant that every component (CPU, GPU, memory controllers, I/O) had to be integrated all together on the same node within the same manufacturing process. This also implied that a defect anywhere within those elements would render the entire chip useless, suddenly making the growing size of a die a major concern as well. To counteract these problems, rather than designing one single massive chip, all of the components that would have been packed in a monolithic die are individually manufactured on their own and integrated together. Each one of those dies are called chiplets and semiconductor leaders like Intel and TSMC are starting to embrace this architecture as a more optimal solution for generating a complete and powerful processor. Feng and Ma (2022) have produced a quantitative cost model demonstrating that partitioning a monolithic chip into multiple chiplets can actually improve the overall output of dies and reduce costs. Beyond performance and economic benefits, chiplet-based architectures have also thwarted much of the criticism towards the environmental footprint of AI, especially its carbon emissions. It has been found that when partitioned moderately, chiplets achieve optimal carbon efficiency and superior sustainability. Although the rise of chiplet architecture is evident, there are still several engineering tradeoffs that exist between monolithic dies and chiplets that make the transition more complex than an economic comparison would suggest. This paper will examine the history of catalysts that forced the shift away from monolithic chips, the functional discrepancies that make chiplets unique, and the real-world applications that ultimately showcase why chiplet architecture could be a promising future for the semiconductor industry.

Literature Review

The study of chiplet architecture has been an ongoing research topic for the past several years, motivated by changing technologies that are exceeding the physical and economic limits of single chip designs. Many papers have now been written covering a range of concerns or unexplored enhancements to illustrate the potential services that chiplets can better provide in the coming years.

Memory system management has been identified as one of the more major components of multi-chip structures and Park et al. (2025) address a consequential issue of memory non-uniformity in their paper. They found that at the intersection of chiplet and GPU memory, accessing data on remote chiplets in turn causes latency and energy consumption that lowers the overall performance of the system. Their main contribution was the introduction of CLAP, a Chiplet-Locality Aware Page Placement tool that deduces the right memory page size for each data structure by analyzing its chiplet-locality or tendency to be accessed by threads of the same chiplet (Park et al., 2025). When evaluated, CLAP improved performances by up to 19.2% when compared to previous paging methods, displaying that if designed together, structuring software and hardware can actually rid the costs of inconsistent memory arrangements.

Power delivery challenges have also garnered much attention due to AI workloads that have added more complexities to chiplet configurations. With the integration of an increasing number of chiplets, efficient power transfer becomes a critical issue when considering how to limit the amount of energy used with every operation. To optimize the power delivery of LLM workloads in chiplet packages, Dong et al. (2024) developed ScalePoM, a framework that selects the most advantageous architectures, workload mappings, and power delivery networks (PDN). Within their framework they also built a chiplet simulator for LLMs that could predict performance, power, latency, and utilization. Through this methodology, they achieved up to 62% saved energy for LLM inferences across a variety of sparsity levels (Dong et al., 2024). Ultimately, their main finding was that power efficiency depended on the grouping and assignment of each chiplet to a specific part of the LLM model, not just the hardware handling the power. Energy consumption becomes even more relevant within the discussion of chiplet limitations as increasingly complex structures start to require even more energy.

One of the greatest benefits and challenges of multi-chip modules come from a chiplet's ability to interconnect with another chiplet that was manufactured from a completely separate factory. The versatility of chiplets is obviously heightened when it can integrate itself with other useful dies like CPUs, GPUs, NPUs, and custom accelerators all in one package. However, the issue arises from the inability to integrate heterogeneous chiplets together without a universal interface shared between all dies. The chiplets need a way to communicate with one another through a standardized interface that includes physical signaling, I/O circuitry and package routing. Lin et al. (2025) provides a solution for this with the development of LEGOSim which introduces a unified integration interface that allows various open-source designed chiplet simulators to join as parallel processes without the need for much modification. They also proposed a synchronization protocol that calls for synchronization to occur only when die-to-die communication actually happens, rather than at a fixed interval. By using on-demand synchronization, the overhead that came from the processes was reduced by up to 99.9% compared to a per-cycle synchronization method. LEGOSim demonstrated that while it can be improved, chiplet compatibility demands intricate solutions for multi-chip architecture to be mass produced by more than one large semiconductor company.

The Historical Stimulus Behind Chiplet Architecture

For about sixty years, the monolithic SoC model represented the foundation of computing progress and the majority of the milestones attributed to the largest semiconductor companies were achieved by trying to pack more and more transistors into one chip. With each new generation of technology, the companies would be gaining the rewards of Moore's Law as transistor counts doubled (Feng et al., 2022). The framework was majorly successful and as time went by transistor sizes shrank, making microchips faster, cheaper, and more energy efficient. System architects counted on each new batch of silicon to deliver significant performance improvements to the chips without having to change its fundamental architectural design.

Complications began to seep through this perfect paradigm, however, as the economic expense that came from silicon usage became more of an issue. When manufacturing a die, the percentage of functional chips that can be produced from a silicon wafer is not a fixed amount. Each wafer contains a certain density of defects (deviations from the crystal structure), and the probability that a die will contain one of these defects only rises with its surface area (Sun et al., 2025). For smaller dies, the amount of chips that can be produced from a wafer is sizable, while for larger dies, even a small density of defects in a wafer can cause a fraction of generated chips to be worthless. This means that the effective cost per every functional chip is a lot higher for dies that are larger in area. As Feng and Ma pointed out, when graphed, the cost per unit area relationship rises rapidly once the die sizes start to reach the limits of what process nodes can economically handle. This ends up making large monolithic dies both expensive to produce and at higher risk of containing possible defects.

In addition to the production costs, the integration of multiple function blocks becomes harder to justify as architects are continuously fitting these units into one chip regardless of their inconsistencies in design. For example, when a company decides to reduce its die size and pack more transistors in it, this optimality does not bode well with logic blocks like analog and I/O circuits since they operate on a very different set of physical standards. Cramming these circuits into a process node with limited voltage tolerances would constrict the range of signals that they can handle and would lower their overall performance tremendously. From an economic viewpoint, advanced process nodes cost a lot of money, which would make having a potential block of circuitry that performs worse than usual an even larger waste of an entire die.

Another indication of a shift in architecture was that the progression of Moore's Law was beginning to stall. The time in between the creation of meaningful process nodes was starting to deviate from its original prediction of two years. Semiconductor companies like Intel and Samsung were continuing to spew out products but every cycle only seemed to show a trivial change in performance with only higher costs. For these reasons, the chiplet-based design started to attract attention as a necessary improvement for the industry. The core implementation was promising, posing the question of why monolithic chips could not be broken into smaller dies that were manufactured and tested individually, and then assembled into one package. The idea of separating each functional block into components that best supported their purpose shined light onto a path where both physical and economic concerns would be mitigated. AMD made the decision to adopt chiplet architecture in 2017, marking an important moment for this transition. In doing so, they were able to offer competitive core totals and heightened performance while providing up to 40% savings in cost compared to traditional designs. The success of this strategy gave chiplets the proper validation in the market and cemented them as the direction the semiconductor field would follow in the future.

Advantages of a Multi-Chip System

At its foundation, a chiplet is a tiny IC (integrated circuit) that is designed to perform a specific, defined function and when interconnected with other chiplets builds a complex component. Among the many advantages chiplets bring, the most key architectural difference that they carry is their reusable IP or intellectual property.

Chiplets support a standardized interface that allows them to communicate with other neighboring dies in a package, mirroring an assembly that is similar to joining Lego pieces together. Each of these pieces, however, take on completely different forms depending on their predetermined type of third party IP. For instance, a chiplet could be designed as a compute chiplet which holds the processor cores and handles the actual arithmetic and logical operations of the system, while another chiplet could be manufactured as a memory chiplet which is dedicated to handling memory and efficient data access (Dong et al., 2024). The workload of a CPU would require a complex piece of silicon for its branching and instruction-level dependencies, but for the workload of a deep learning inference which has far more predictable memory access patterns, all of those intricacies would be wasted. Since each unit carries different purposes, the performance of every functional piece can be maximized by configuring them with the best fitting process node.

Another unique aspect of chiplet functionality is its ability to tailor to the specific bandwidth and latency needs of the memory subsystem. In 2015, mass production of High Bandwidth Memory was introduced which stacks multiple DRAM dies vertically and places memory closer to the processors to offer faster data rates. With the introduction of HBM, memory bandwidth increased heavily compared to conventional DDR DRAM, but ultimately sacrificed cheaper costs of production and simplicity. Integrating HBM with a monolithic SoC also became an arduous challenge due to its physical size, contrasting the current purpose and trend of monolithic dies becoming smaller and tighter in design. Thermal management presented additional threats as both HBM stacks and SoCs generate lots of heat, creating hotspots and much more complicated solutions to cooling down the entire system. Issues escalate here when the processors exceed safe temperatures and end up slowing down the operating speed to lower the overall heat output (Park et al., 2025). Lastly, the combination of HBM stacks and monolithic SoCs' high prices created a drastically crippling yield if defects were found in either component. By instead using a chiplet architecture, HBM can now be integrated adjacent to the compute chiplets rather than within the same die. One method of doing so is through a large silicon interposer that sits underneath the chiplets and the HBM stack that provides a route for the chiplets to communicate with HBM. The same required bandwidth can now be transferred but at a fraction of the energy cost of a conventional memory bus. Distance plays a significant factor in why, due to the fact that signals need to travel much less in a chiplet system at far lower voltages.

A more practical benefit of chiplet architecture stems from being able to test each die individually before they are assembled and packaged all together. In a singular chip, testing is straightforward in the sense that it is tested in its entirety once and is discarded if there are any deficiencies. However, the cost of that failure is an entire die. In a chiplet-based system, each chip component is carefully tested one at a time and once all functional checks are passed, they are moved onto the packaging stage. Chiplets undergo burn-in testing which subjects semiconductors to high temperatures and voltages to spot any failures early on before they get shipped. Not only does this improve the overall yield that is possible from the total silicon wafer, but also reduces the risk of having an unstable product once packaged together.

Power efficiency adds additional value to chiplet architecture by reducing the net power consumption of the die. Smaller modules are fundamentally more efficient in power usage because they contain shorter wire lengths, reducing the capacitance and electrical energy needed to switch states. The node optimization feature explained previously presents an even more favorable design for energy management, since all of the computational chiplets can be placed on advanced process nodes while less performance-heavy blocks can be kept on mature nodes where they would operate more efficiently (Dong et al., 2024). Chiplets can require the addition of multiple voltage domains to assist with managing power delivery more effectively. A voltage domain regulates a circuit's power supply so that it is most ideal for that tile's functionality. In the case of a chiplet that is not actively processing, current to that die can be shut off without affecting the usage of neighboring chiplets, an operation that is much more difficult to achieve when power domains are isolated from each other. Power management develops into an increasingly valuable factor for multi-chip designs as the counterargument of thermal constraints becomes a much more prominent issue.

At times, an overlooked source of energy consumption in recent computing systems is in data movement. The fraction of energy that is needed to transfer data between processing units and other components when computing operations is only increasing as processor performance has accelerated. Similar to how wire distance was affecting power utilization, the physical proximity of a compute chiplet and a memory chiplet that share the same silicon interposer also plays a role in how much energy is being used to send signals back and forth. When packaged together the space between is reduced, translating to lower signal energy and less of a swing in voltage when bits are being transferred. The cost of fetching a byte of data from DRAM to a processor is also not an insignificant amount. By reducing this energy cost, workloads that involve copious amounts of data like machine learning and database processing will benefit from an improved, energy efficient system.

Chiplets provide a vast advantage over singular dies when it comes to the integration of diverse technologies that would not be as compatible in a monolithic design. The leverage gained can be illustrated through the fusion of chiplet modules and silicon photonics. Silicon photonics is a recent development in technology that uses light instead of electrical signals to transfer data through silicon chips, allowing for a much faster and energy-efficient form of communication. With the imminent demand of AI and cloud-based workloads, massive amounts of bandwidth and lowered latency is required to satisfy these needs. Compared to the electrical consumption of copper interconnects, photons do not generate resistive heat and can travel long distances without reducing the data moved. This does seem highly attractive for communication between processors and memory but the manufacturing process of photonics involves differing materials and structures which are difficult to combine with traditional semiconductor construction. That complexity is less of an issue in a chiplet system due to die-to-die interconnects that bridge each unit together. Since there is no need for all components to be built on the same process, a silicon photonics chiplet can be developed on a node most optimal for all the lasers, modulators and photodetectors necessary for a photonic link. The interconnection between dies handles the boundary between electrical and optical domains through the silicon photonic modulator, taking a continuous laser beam and altering the phase or intensity based on the electrical input. Furthermore, by applying the extensive chip testing mentioned previously to the individual optical components, any performance failures can be modified before assembly takes place.

Chiplet architecture is not attempting to tailor its structural design for every novel piece of hardware, but is instead removing the necessity for manufacturing compatibility to even be considered as a factor of production. As long as the technology can be produced into a die that is adaptable to the standardized chiplet interface, its integration into the packaged system should be feasible. The advantages described present a compelling argument towards the universal solution that is multi-chip modules, offering competitive yields, lower costs, and most importantly greater flexibility. However, it would be incorrect not to mention that the design choices that give chiplets this edge on monolithic chips also introduce a set of engineering challenges that sway the immediate decision to stamp chiplets as our most promising semiconductor solution of the future.

Limitations and Challenges of Chiplet Systems

A major concern in any processor design is how overheating will be managed, but in chiplet architecture, thermal control becomes a much more complex issue compared to monolithic designs. In a singular die, heat is generated across the entire surface and is conducted through one path to the cooling solution. The behavior of the heat is typically predictable and is transferred efficiently from the chip's surface through the heat spreader. In a package of chiplets, numerous dies of varying sizes and power densities are clumped very close together and when run at full load, significant amounts of heat per unit area can be generated. Varying functionality between chiplets makes it even harder to manage the heat consistently since one die may be producing more thermal energy than another. Since neighboring dies are in close proximity with one another, hot spots can spread through heat conduction and increase the overall temperature of other chips. This implies that power budgeting needs to be carefully managed across all dies, causing performance limitations on chiplets that emit the most heat.

Although chiplet-based systems provide a lot more freedom to integrate advanced new solutions, the design also involves more sophisticated engineering strategies that chip designers need to hurdle. Breaking one big chip into smaller components may seem like the same amount of work is being completed, but there are now new tools required to interlink multiple dies, process nodes and layers together, all while preserving the interactions existing between them. Due to 2.5D/3D multi-chiplet designs, advanced packaging technology needs to be enforced, introducing mechanical form factors and signal/power integrity management. Signal analysis must handle the electrical aspects of die connections while power analysis needs to handle the voltage drop across the package delivery network (Lin et al., 2025). Attempting to assemble all of these packaging layers together is a daunting challenge and will continue to require significant attention on the engineering end.

Chiplets' Involvement in the Growth of AI and Sustainable Computing

Modern artificial intelligence is placing increasingly more difficult demands on computing hardware each year that the semiconductor industry must constantly adjust to. The process of training an LLM like GPT-4 requires a constant operation of thousands of processors over long periods of time to move extensive sizes of data through memory systems. It will become harder for monolithic dies to accommodate the essential bandwidth AI workloads need at an acceptable yield and cost, which will cause an adoption of chiplet products. The matrix multiplication operations that dominate the computational costs of neural networks benefit greatly from parallelism which can be achieved through the assembly of multiple dies. The plausibility of this idea has been further proven through AMD and their Instinct MI300X Multi-chiplet Accelerator which has been designed for generative AI and high-performance computing. The MI300X combines multiple compute dies and HBM stacks onto a single silicon interposer, delivering 192 gigabytes of HBM3 memory for large model inference. By packaging the memory stacks on the same interposer as the compute dies, chiplet architecture discards the bottleneck that occurs when the connection between the processor and external memory cannot supply data quickly enough. NVIDIA has also released their own approach to chiplet integration with the Blackwell architecture in 2024, which links two dies via a chip-to-chip interconnect into a single GPU. The model is described to be packed with 208 billion transistors, which is a count that would be much too large for a single monolithic die using any process node available right now. Both of these systems have become stimulus for AI technology and without the ability of chiplets, a lot of the hardware that powers today's machine learning systems would not exist.

Beyond the achievable energy and power efficiency mentioned earlier, there are environmental repercussions involving carbon emissions that chiplet architecture aids heavily. Sun et al. (2025) singles out the manufacturing phase of semiconductor production to be the stage that generates a significant amount of carbon, a point that is often ignored in discussions. A rigorous analysis on the carbon footprint of chiplet-based AI accelerators was carried out and researchers found that only in the case of large functional areas and moderate chiplet counts, was the chiplet model superior in carbon production (Sun et al., 2025). This finding demonstrates the tension that results from separating chip modules. More specifically, the contemplation between balancing the manufacturing advantages and the additional carbon overhead that comes with the decomposition. On one hand, even if a chiplet design has a low carbon yield per-die, the total footprint can still end up being greater than a monolithic design if the number of chiplets is high. Conversely, it was proven by Sun et al. that an optimized packaging system designed to combine a tolerable amount of chiplets achieves lower carbon emissions than any other single die alternative.

The pivotal consideration here is balance and being able to achieve the required purpose of a packaged SoC, while keeping the number of chiplets to the lowest quantity it needs to be. A chiplet's ability to be reused across multiple generations of products without the need for a redesign also limits the recurring carbon cost that would have existed if that chip was to be made at every iteration. Artificial intelligence and the carbon generated by its hardware will only continue to rise, thus making the idea of regulating die counts and producing sustainable systems through chiplets a much more promising proposition.

Conclusion

What emerges from this back and forth discussion between the convenience of chiplets and their potential downfall, is a presentation of the powerful and demanding value that chiplets carry over monolithic designs. The anatomy of their systems has been explored in detail and the benefits that the separation of chips brings cannot be denied. The semiconductor industry has clearly shown that the tradeoff between using single die SoCs and multi chip architectures is worthwhile and have even committed to using chiplets for their most demanding hardware. The concept of an open chiplet marketplace where specialized dies from different vendors can connect seamlessly through a universal standard is still an ambitious goal, but the prospect of it occurring highlights a hopeful direction for the endurance of chiplets in the long run.

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